All 441 minimum stopping supports of the source B1 matrix

24 October 2026 · Expanded verified checkpoint · L-F16 follow-up

Abstract. We classify every size-six stopping support of the exact 882-column B1 X-check matrix displayed by Abiad and Castriota. They are precisely the 441 distinct companion Z-check row supports, forming one free orbit of Z/7×Z/63 under coordinate translations. No connectedness, elementary-support or even-syndrome restriction is assumed. A generic all-branch classifier, a two-root certificate evaluated in Lean’s kernel, unique-center and cardinality proofs, and the actual binary matrix bridge establish the result. This expands the preceding exact-distance family paper, reproduced and updated below; its spectral comparison remains unchanged.

Question, source and answer

Question / source	Answer / evidence
Original parameter question: Abiad-Castriota2026, §6	Answered negatively in the separate source-answer paper.
Project-derived B1 minimum-support classification	All441 size-six base supports are unique companion Z-check rows, one free product-group orbit; checked complete classification.
Earlier project-derived cover-family result	Stopping distance6 for every positive63-multiple lift; retained below, not a classification of all covers.

Documentation revision: full Lean URLs, actual proof excerpts, dependency guide and complete calendar-delta ledger are supplied in Appendices A-C. No existing proof or source-question count changes; the additional dense-source bridge is explicitly identified.

1. Complete classification of the source B1 matrix

Abiad and Castriota [1], arXiv:2607.13666v1, 15 July 2026, Example 3.2 and §6, motivate tighter stopping bounds for QC-GHP codes. Our preceding checked theorem gives stopping distance six for B1 and positive 63-multiple cyclic covers. The user-requested follow-up here asks for every minimum support of the BASE matrix. It is our project-derived restricted target, rather than another author-posed canonical question.

Theorem 1 (complete minimum-support classification). Use lift length 63 and the matrices defined in Section 2 below. With $G=Z/7\times Z/63$ and left/right variable coordinates, put

$$R(j,f)=\{\mathrm{left}(j,f),\ \mathrm{left}(j,f-1),\ \mathrm{left}(j,f-6),\ \mathrm{right}(j,f+27),\ \mathrm{right}(j+1,f),\ \mathrm{right}(j+2,f+54)\}.$$

For every finite variable support T, the following are equivalent: T is a nonempty H_X stopping set with $|T|=6$; there is a unique $c\in G$ with $T=R(c)$; there is a unique $c\in G$ with T equal to the support of binary H_Z row c. There are exactly 441 such supports. All are translates of $R(0,0)$, and no nonzero coordinate translation fixes it. Each has three variables per side, and every check has induced degree zero or two.

The translation group is the PRODUCT $Z/7\times Z/63$, not a cyclic group of order 441. Distinct row supports do not imply linearly independent generators. This theorem classifies only the 882-variable base matrix. The family stopping-distance theorem below remains valid for every positive 63-multiple lift, but a complete classification of those larger covers is not asserted.

Proof, first step: a sound exhaustive classifier. Maintain a duplicate-free partial support $S\subseteq T$ and a fuel bound $|T|\le|S|+\mathrm{fuel}$. At positive fuel, find a degree-one check and require every extension through an unselected neighbor to pass. If no such check exists, the certificate rejects. The stopping property ensures that one child remains inside T. At fuel zero, containment and cardinality force $S=T$; a remaining degree-one check excludes this leaf, and otherwise its specified row form must be checked. Induction proves that a true certificate implies the requested form for every stopping completion. Premature stopping is rejected explicitly, so no hidden minimality premise is used.

1. Complete classification: proof and checked evidence

Second step: two normalized roots. Both coordinate translations preserve the actual incidence relation. Translating any selected variable to (side,0,0) reduces all supports to two root cases. The good-leaf test checks length six and containment in R(c) for one of six centers: (0,0), (0,1), (0,6), (0,−27), (−1,0), (−2,−54). Duplicate-freeness and |R(c)|=6 turn containment into equality. Ten kernel-checked branch certificates combine by a proved generic rule into the two-root certificate with five additions. Translating back gives existence of a center for every T.

Third step: uniqueness and count. A second kernel certificate checks that R(c)⊆R(0) forces c=0. If R(c)=R(d), translation by −d gives R(c−d)=R(0), hence c=d. The complete minimum-support family is proved equal to the image of all centers; the injective-image cardinality formula gives 441. The same translation identity gives one free G-orbit. Direct transport of the base profile proves that every minimum support has three variables per side and induced check degrees zero or two. These properties are conclusions, not input restrictions.

Fourth step: actual binary source bridge. Simultaneously translating row and variable coordinates preserves the explicit natural H_Z matrix. The known R(0) row identity gives $R(c)=\{v:H_Z(c,v)=1\}$. All coefficients are proved zero or one before casting to F_2. The final theorem is an equivalence with a UNIQUE row of the same binary HX2/HZ2 source matrices used in the earlier family theorem. It assumes neither the classification nor its count.

New proof / bridge	Checked Lean declaration
All-branch classifier soundness	ClassificationSearch.lean: classifiesList_sound
Good forms; kernel certificate; arbitrary supports	ClassificationData.lean: good_sound; ClassificationBranches.lean: branch certificates; ClassificationKernel.lean: normalized_classification_certificate; Classification.lean: size_six_classification
Distinct centers, 441 count, free orbit	ClassificationCertificate.lean: rowSupport_injective, unique_center, minimumSupports_card, single_translation_orbit, translation_free
Actual binary matrix / Z-check row equivalence	ClassificationCertificate.lean: HZ_shift, rowSupport_HZ, HZ_binary, binary_source_classification
Three variables per side; elementary profile	ClassificationCertificate.lean: minimum_structure, binary_source_structure

Independent corroboration searches all 882 roots, visiting 3,445,092 nodes and finding exactly the 441 companion-row supports. It uses a different check ordering from Lean. This is not a proof premise. Known stabilizer stopping sets [4-6] supply the upper mechanism; the present result additionally proves exhaustiveness at size six. Rooted enumeration itself is classical [3]. No global priority claim is inferred from the bounded literature/GitHub/X audit.

New source files, complete signatures, an all-45-declaration axiom audit, inherited-source hashes and inventory validators accompany this expanded checkpoint. The source_complete_certificate theorem combines the count and the equivalence with ALL actual binary size-six supports. All new claims are checked in Lean; only standard axioms are permitted. Source PDFs remain outside the archive. No logical-dimension, logical-distance or decoding-performance conclusion is drawn.

2. Check family, source question and scope

Abiad and Castriota [1], 15 July 2026, arXiv:2607.13666v1, Proposition 5.1, Remark 5.2 and Section 6, ask for QC-GHP parameters making their spectral stopping bound informative and for tighter bounds. The preceding research block establishes a scalar-factor obstruction to the first route. This paper answers our narrower, incidence-aware follow-up: can actual incidences certify the exact stopping distance of the running B1 matrix and its positive 63-multiple covers?

Theorem 2 (preceding family result). Let t be a positive integer, $\ell=63t$, and let P(s) have entries $P(s)_{i,j}=1$ exactly when $i=j+s$ in $\mathbb{Z}/\ell$. Let S be the forward cyclic shift of order 7. Over $\mathbb{F}_2[x]/(x^{\ell}-1)$, put

$$A=x^{27}I_7+S+x^{54}S^2, \quad b=1+x+x^6,$$
$$H_X=[\tilde{A} \mid I_7\otimes\tilde{B}^T], \quad H_Z=[I_7\otimes\tilde{B} \mid \tilde{A}^T].$$

Then $H_XH_Z^T=0$ over $\mathbb{F}_2$. Each matrix has 882t columns and 441t rows; H_X has column weight 3 and row weight 6. Every nonempty H_X stopping support has size at least six, and an H_Z row is a size-six stopping support. Thus the H_X stopping distance is six. For the real expanded matrix, $\sigma_2(H_X)^2\geq 12$, counting Gram eigenvalues in descending order with multiplicity.

Scope. These are CSS check matrices. We do not establish logical dimension, logical distance, rate or decoding performance for the covers. A stabilizer support is not a logical operator. At t=1 the displayed matrices are the source's Example 3.2; its reported [[882,24,d]], $18\leq d\leq 24$ parameters are cited, not reproved here. The family does not satisfy Proposition 5.1's spectral inequality.

3. A sound lower certificate on the base

For a binary incidence relation E(v,c), a stopping support is a nonempty finite variable set T such that no check has exactly one neighbor in T. No elementary, connected-support, or even-syndrome hypothesis is imposed. With checks (j,f) in $Z/7\times Z/\ell$ and variables (side,j,f), the source matrices give precisely these column lists:

left(j,f): (j,f+27), (j+1,f), (j+2,f+54);
right(j,f): (j,f), (j,f−1), (j,f−6).

Lemma 2 (extension soundness). Suppose S is a duplicate-free list contained in a stopping support T. If a check has one neighbor in S, it has a second neighbor in T outside S. Branching over every other neighbor of that check therefore preserves at least one branch contained in T. If $|T|\le|S|+f$, f further additions suffice to reach a partial set with no degree-one check. Consequently rejection of every branch excludes all such stopping supersets.

Proof. If no check has degree one in S, the test accepts. Otherwise the stopping property supplies the extra neighbor. Append that neighbor, retain duplicate-freeness and containment, reduce the fuel by one, and apply induction. Completeness of the explicit row list and the check picker’s degree-one condition are proved separately. An accepting branch is not used to infer a lower bound. □

Lemma 3 (two roots). Simultaneously translating both cyclic coordinates of all variables and checks preserves incidence, support cardinality, and the stopping property. Every selected variable translates to (side,0,0). Thus it suffices to exclude small stopping supports containing either of the two side representatives.

Proof. Each entry in the displayed incidence lists is affine with coefficient one in the translated coordinates. Translation is a bijection on variables and on checks. Image cardinalities and the selected-neighbor cardinality at every check are therefore preserved. Nonemptiness supplies the selected variable. □

Base certificate. At $\ell=63$ the checked Boolean extension test rejects both roots with four further additions. Lemmas 2-3 exclude every stopping support of size at most five. This is an exact finite certificate, evaluated by **decide +kernel**; neither external enumeration nor native evaluation is a proof premise.

The independent Python reconstruction also searched all 882 base roots, using 688,842 search nodes, without finding a support of size at most five. This corroborates the model and implementation. The Lean certificate and its soundness theorem supply the proof.

```
normalized_roots_reject :
∀ b : Bool, survivesList pickList row 4 [(b,0,0)] = false


running_lower_six (T : Finset Var) (hT : Stopping E T) : 6 ≤ T.card
```

The search strategy is classical rooted narrowing and branch-and-bound; Wang-Kulkarni-Poor [3] are relevant prior art. The contribution here is the checked instance, exact source translation and cover extension, rather than a new general enumeration algorithm.

4. Cover transfer, upper witness and CSS bridge

Lemma 4 (local-cover transfer). Let variable and check maps take a graph to a base graph. Assume: incidences map to incidences; every base check incident to the image of a selected variable has an incident lift at that variable; and the variable map is injective on each check's neighbor set. Then the image of every stopping support is a stopping support, with cardinality at most the original support.

Proof. The image is nonempty. If a base check meets it, choose a selected variable over that meeting and lift the check. The original stopping property provides a different selected neighbor at the lifted check. Local injectivity makes their images distinct; incidence preservation puts both at the base check. Hence the base check cannot have selected degree one. The finite-image cardinality inequality gives the size claim. □

For $\ell=63t$, reduction $\mathbb{Z}/\ell\to\mathbb{Z}/63$ maps both explicit column lists onto their base lists. The six neighbors of any lifted check map to six distinct base neighbors. The reduction homomorphism, incidence preservation, lifting and local injectivity are all proved in Lean. Lemma 4 and the base lower certificate give the lower bound six for every positive t; this is not extrapolation from sampled lift lengths.

Upper witness. For every such lift the following support has six distinct variables:

left(0,0), left(0,−1), left(0,−6),
right(0,27), right(1,0), right(2,54).

Every incidence of a selected variable has an explicitly checked selected partner on the other side, so no check has selected degree one. The support is exactly H_Z row (0,0). For the base, kernel reduction checks that every induced check degree is zero or two. These facts give the upper bound six. Combining it with the lower bound proves Theorem 2's exact stopping distance.

Actual matrix bridge. The formalization first defines the displayed matrices over the naturals. Duplicate-free column lists imply their entries are zero or one, and their count predicates are proved equivalent to the incidence lists. Casting these entries to F_2 preserves precisely the same graph. Separate counting lemmas prove the dimensions and H_X degrees. Thus the stopping theorem concerns the actual binary matrices, not only an abstract graph model.

CSS proof. On the finite abelian product $\mathbb{Z}/7\times\mathbb{Z}/\ell$, $\tilde{A}$ and $D=I_7\otimes B$ are circulant matrices. Circulant commutation gives $\tilde{A}D^T=D^T\tilde{A}$. The expanded binary product is $\tilde{A}D^T+D^T\tilde{A}$, which vanishes in characteristic two. All identities, transpose conventions and casts are checked for the displayed matrices.

The existence of stabilizer stopping supports is known in quantum erasure decoding [4,5], with the mechanism attributed there to Connolly et al. [6]. The witness is an instance of that mechanism. The exact lower bound and the cover comparison are separate claims; the witness alone does not establish them.

5. Spectral comparison and known stopping sets

Theorem 5. For every positive lift in Theorem 2, the actual real matrix has $\sigma_2^2 \geq 12$.

Proof. Consider the two-dimensional plane of row vectors constant on fibers, supported at block rows 0 and 3, with real coefficients a and b. Its squared norm is $\ell(a^2+b^2)$. Direct sums over the expanded matrix show that the left block contributes $3\ell(a^2+b^2)$ and the right block contributes $9\ell(a^2+b^2)$. The energy is therefore 12 times the squared norm. The linear functional giving the top eigenvector coefficient has a nonzero kernel on this plane. A nonzero vector in that kernel has Rayleigh quotient at most the second Gram eigenvalue, including multiplicity. Hence that eigenvalue is at least 12. All finite sums and the spectral reduction are checked in Lean. □

Proposition 5.1 of [1] requires $\sigma_2^2 < \gamma/2$; here $\gamma=3$ and $12>3/2$, so the condition fails. In the preceding block, inspection of the exact radicals and the separate exclusion of the small-size branch also showed why satisfying the spectral inequality alone need not give a useful stopping-distance guarantee. We do not apply that proposition outside its hypotheses.

For comparison, Zhu-Chugg [2], Corollary 6, equations (41)-(42), give the following two regular-graph stopping bounds, with N variables, column degree γ, row degree k and second Gram eigenvalue η:

$$N(2\gamma-\eta)/(\gamma k-\eta),$$
$$2N(2\gamma+k-2-\eta)/(k(\gamma k-\eta)).$$

Their strict spectral-gap regime is $\eta<\gamma k$, with the connected regular setting supplying the unique top eigenvalue. These formulas have no separate small-support branch. They use a different threshold from [1]. In our family $N=14\ell$, $\gamma=3$, $k=6$ and $\eta\geq 12$. If $\eta<18$, both denominators are positive but the numerators have factors $6-\eta$ and $10-\eta$, both negative. Thus both expressions are negative. If the strict gap fails, the displayed bounds are outside that regime. The two cases require no numerically inferred connectivity or spectral gap. This comparison does not dismiss Zhu-Chugg's broader multi-eigenvector or irregular analysis.

As an illustration only, independent dense reconstruction at $\ell=63$ gives $\eta\approx 15.89294145$, and the two formula values ≈ -4141.1162 and ≈ -822.2481. Floating-point values do not enter the proof.

Prabhu et al. [4] distinguish stabilizer, horizontal and vertical stopping supports and study erasure decoders. Their constituent expansion conditions are not spectral conditions on expanded H_X. Yao et al. [5] explicitly simulate B1 and study post-peeling cluster decoding. The inspected full texts do not supply the lower certificate proved here. Plain peeling obstructions and quantum-aware correctability are different questions.

The 2026 HGP stopping-set paper [7] remains a full-text access gap; its abstract is insufficient for a complete theorem comparison. The public code and X searches were bounded; they support no global absence or priority claim. The detailed search and source-screening ledgers accompany the proof bundle.

6. Family proof evidence and reproducibility

The aggregate theorem **Incidence.Family.positive_family** quantifies over every natural t with [NeZero t] and combines dimensions, H_X degrees, actual binary CSS commutation, the universal lower bound for stopping supports and an existential size-six witness. The source B1 consumer specializes it to t=1. The spectral theorem uses the descending ordered Gram spectrum of the same expanded real matrix.

Claim / proof	Lean source and main declaration
Stopping definition; extension; translation	IncidenceCore.lean: stopping_extension, shift_incidence, stopping_image
List-search soundness and picker	IncidenceList.lean: survivesList_of_stopping, pickList_singleton
Kernel certificate and base lower bound	Incidence.lean: normalized_roots_reject, running_lower_six
Abstract cover theorem	IncidenceCover.lean: stopping_cover_image
Concrete cover; six-element support	IncidenceFamilyCore.lean: maps, lifts, local_inj, six_card, six_stopping
Base source matrices and profile	IncidenceSource.lean: HX_incidence, six_profile, six_is_HZ_row
Family source/binary and companion-row bridge	IncidenceFamilySource.lean: HX_binary, six_is_HZ_row
Cover lower bound / exact natural matrix	IncidenceFamily.lean: lower_six, source_matrix_exact_six
Actual binary CSS and incidence	IncidenceCSS.lean: css, HX2_incidence
Degrees and dimensions	IncidenceDegrees.lean: column_weight, row_weight, dimensions
Expanded spectrum and Zhu-Chugg comparison	IncidenceSpectral.lean: plane_energy, second_lower_twelve, zhu_chugg_bounds_negative
Combined positive family and source consumer	IncidenceCertificate.lean: positive_family, source_B1_stopping_distance_six

Pinned environment: Lean 4.36.0-rc1; mathlib commit 5f7b4198d22a9c82e1c698a379334fe128095c82. The standalone project vendors the parent Spectral.lean unchanged, SHA-256 a3e712b2a206bd7743408ecc122e66e7d980dbd4689c6d49977536dfe9bb469d. It is inherited infrastructure, not a newly counted result. The preceding family aggregate build completed successfully (5,494 jobs). The accompanying audit checks every named declaration against the standard axiom allowlist. Exact commands, signatures, build log and hashes are in the archive.

Reproduce the expanded checkpoint: **lake build ClassificationCertificate**; then **lake env lean AuditClassification.lean**. Family modules are under dependencies/. README.txt explains cache setup. Python cross-checks reconstruct the integer matrix independently, exhaust all 882 base roots, and test lift lengths 63t for t=1,...,8. They corroborate the formal result; the universal theorem is established in Lean.

The checked proof sources and paper have been published to GitHub. Appendices A-B give actual declarations, dependencies and complete immutable file URLs; verification.json records current hashes. Published folder: github.com/alectionary/vibe-math, formalization/stopping-bound/classification/. Complete file links and inherited-source hashes are in PROOF-LINKS.html and verification.json inside the source archive. The X announcement remains an unposted draft.

7. Chronology, inventory and limitations

Source	Relevant date / inspected milestone
Tanner classical spectral ingredient	Inspected manuscript 26 July 2000; later 2001 publication not independently inspected
Zhu-Chugg	28 September 2005 conference metadata; December 2007 journal inspected (month precision)
Wang-Kulkarni-Poor	19 May 2006 preprint; inspected journal version 22 April 2009 / May 2009
Panteleev-Kalachev construction	4 April 2019 preprint; 12 November 2021 v3 inspected in parent block
Yao-Gokduman-Pfister	11 December 2024 v1 inspected
Prabhu et al.	30 April 2025 v1; 29 May 2025 v2 inspected
Abiad-Castriota question [1]	15 July 2026 v1
This checkpoint	24 October 2026; 101 calendar days after [1]

The 101-day difference is date arithmetic, not an assertion about how long a problem remained unsolved. The original investigation began at 19:49 UTC; the incidence follow-up began at 20:48 UTC and the classification follow-up at 21:38 UTC on 24 October. Verification and manuscript checkpoint times are recorded separately in verification.json. Classical ingredients' ages do not measure the age of this exact question.

The preceding incidence block screened three additional accessible full texts for open questions. Five broad historical directions entered the shared inventory with current resolution and Lean status unassessed; one 2026 paper retains unknown question count due to missing full text. The cohort moves from 262 to 267 canonical entries, with 135 eligible targets and four completed canonical Lean targets unchanged. This restricted project-derived follow-up supports L-F16; it is not another completed author-posed target. The separate original parameter question is now recorded as answered negatively (5/135 completed canonical Lean targets); the four-target counts in earlier dated checkpoints are historical.

Question answered. All 441 size-six supports of the base B1 X-check matrix are unique companion Z-check rows, in one free coordinate-translation orbit. The preceding exact-distance-six cover-family result is retained. **Question still open.** A broadly applicable replacement stopping bound and the logical ranks/distances of these covers remain open. General matrix-factor extensions are outside the original proposition's stated model, rather than unhandled cases of its answered parameter question. **Classification update.** Theorem 1 completes the base minimum-support classification. This block adds no canonical entries, eligible targets, paper records, full-text screens or canonical completions. **Next proposed target (follow-up).** Classify minimum supports in every positive 63-multiple cover, using the checked projection and base structure. This extension has not been proved here.

References

[1] A. Abiad and N. Castriota. Spectral and Additive Combinatorial Methods for Cycles and Absorbing Sets in Lifted-Product Quantum LDPC Codes. arXiv:2607.13666v1 (15 July 2026), Example 3.2, Proposition 5.1, Remark 5.2, §6. https://arxiv.org/html/2607.13666v1

[2] M. Zhu and K. M. Chugg. Bounds on the Expansion Properties of Tanner Graphs. IEEE Transactions on Information Theory 53(12), 4834-4840 (December 2007), Corollary 6. https://hal.usc.edu/chugg/docs/pubs/zhu2007bounds.pdf

[3] C.-C. Wang, S. R. Kulkarni and H. V. Poor. Finding All Small Error-Prone Substructures in LDPC Codes. IEEE Transactions on Information Theory 55(5), 1976-1999 (May 2009). DOI:10.1109/TIT.2009.2015993.

[4] Prabhu et al. arXiv:2504.21845v2 (29 May 2025), §III.B-D and §VII. https://arxiv.org/abs/2504.21845v2

[5] Yao, Gokduman and Pfister. arXiv:2412.08817v1 (11 December 2024), §III and Figure 7. https://arxiv.org/abs/2412.08817v1

[6] Connolly et al. Quantum 8, 1450 (27 August 2024). DOI:10.22331/q-2024-08-27-1450. Metadata and cited stabilizer mechanism only; full text not independently screened in this follow-up.

[7] Prabhu, Gayathri and Kumar. On the Analysis of Stopping Sets Associated with the Quantum Hypergraph Product Code. ISIT 2026. DOI:10.1109/ISIT62367.2026.11654127. Named full-text access gap.

Appendix A. Actual checked Lean and dependencies

This is a documentation revision of an already checked result, not a new research target. The declaration and proof below are extracted verbatim from the committed Lean source. The earlier proof-to-Lean tables identify the individual mathematical claims; Appendix B gives full current and immutable URLs for every research and audit module in this artifact, including inherited dependencies.

```
theorem source_complete_certificate :
    minimumSupports.card = 441 ∧
    (∀ T : Finset Var, T ∈ minimumSupports ↔
      Stopping (fun v c => Incidence.Family.HX2 (n := 63) c v = 1) T ∧ T.card = 6) ∧
    (∀ T : Finset Var,
      (Stopping (fun v c => Incidence.Family.HX2 (n := 63) c v = 1) T ∧ T.card = 6) ↔
      ∃! c : Check, T = Finset.univ.filter (fun v => Incidence.Family.HZ2 (n := 63) c v = 1)) := by
  exact ⟨minimumSupports_card,mem_minimumSupports_binary,binary_source_classification⟩
```

Dependency guide

Inherited incidence/CSS/family sources -> ClassificationData. ClassificationSearch soundness + ten ClassificationBranches certificates -> ClassificationKernel -> Classification arbitrary supports. ClassificationCertificate adds unique center, count441, free orbit and actual binary-matrix bridge.

Translation and boundary check: source binary expanded incidence is the real zero-one matrix used for its Gram spectrum, with descending eigenvalues counted with multiplicity; stopping supports are nonempty and permit stabilizer supports. The source roots retain the separate small-size branch. The cover and classification interfaces instead use the actual binary HX2/HZ2 matrices; base-only classification and positive-cover distance are distinct statements. These definitions and bridges, rather than a numerically assumed spectral envelope, carry the translation.

The imported research modules are included in the published source tree. Only standard mathlib axioms appear in the recorded audits. Source interpretation and prior-art comparisons remain documentary judgments; a successful Lean build alone does not certify those judgments. The only added formal work is the explicitly identified dense-source bridge in the parameter-answer artifact; the earlier universal, family and classification proofs remain unchanged.

Appendix B. Complete Lean file URLs

Every URL is printed in full and is clickable. Current links follow master; immutable links identify the exact checked file at revision b7628643bb256bf9c1a319f16f6e19608116cff6. Audit programs and vendored dependencies are included rather than hidden behind folder links.

classification/AuditClassification.lean - axiom/signature audit

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/AuditClassification.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/AuditClassification.lean

classification/Classification.lean - checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/Classification.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/Classification.lean

classification/ClassificationBranches.lean - checked proof module

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationBranches.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationBranches.lean

classification/ClassificationCertificate.lean - checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationCertificate.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationCertificate.lean

classification/ClassificationData.lean - checked proof module

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationData.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationData.lean

classification/ClassificationKernel.lean - checked proof module

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationKernel.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationKernel.lean

classification/ClassificationSearch.lean - checked proof module

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationSearch.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationSearch.lean

classification/dependencies/Incidence.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/Incidence.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/Incidence.lean

Appendix B. Complete Lean file URLs (continued)

classification/dependencies/IncidenceCSS.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCSS.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCSS.lean

classification/dependencies/IncidenceCertificate.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCertificate.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCertificate.lean

classification/dependencies/IncidenceCore.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCore.lean

classification/dependencies/IncidenceCover.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCover.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCover.lean

classification/dependencies/IncidenceDegrees.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceDegrees.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceDegrees.lean

classification/dependencies/IncidenceFamily.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceFamily.lean

classification/dependencies/IncidenceFamilyCore.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceFamilyCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceFamilyCore.lean

classification/dependencies/IncidenceFamilySource.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceFamilySource.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceFamilySource.lean

Appendix B. Complete Lean file URLs (continued)

classification/dependencies/IncidenceList.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceList.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceList.lean

classification/dependencies/IncidenceSource.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceSource.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceSource.lean

classification/dependencies/IncidenceSpectral.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceSpectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceSpectral.lean

classification/dependencies/Spectral.lean - inherited proof dependency

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/Spectral.lean

Appendix C. Calendar deltas for all compared sources

Civil calendar days to 2026-10-24; exact dates give exact differences. Month/year precision yields inclusive minimum/maximum day ranges; dates are not invented. Date ranges are clipped to the revision because these records already existed then. Source age is not unresolved-problem duration.

This consolidated ledger covers the original parameter result and both follow-ups. Rows explicitly distinguish full-text mathematical comparisons, metadata-only or unavailable papers, discovery provenance, software snapshots and inspected X posts. Rechecking dates did not add a paper or a new full-text question screen. Earlier table dates remain valid; this table supplies the previously missing arithmetic and additional sources.

Source / role	Milestone date and elapsed calendar days	Access / precision limit
F16: Abiad-Castriota: motivating parameter question Original question; Prop.5.1 / Remark5.2 / §6	first and inspected v1: 2026-07-15; 101 days (day precision)	inspected full text
F68: Tanner: Minimum Distance Bounds by Graph Analysis Classical spectral ingredient	earliest inspected manuscript: 2000-07-26; 9586 days (day precision) journal milestone: 2001; 9063-9427 days (year precision)	inspected full text
F67: Zhu-Chugg: Bounds on the Expansion Properties of Tanner Graphs Classical stopping bound; exact Cor.6 comparison	conference predecessor metadata: 2005-09-28; 7696 days (day precision) inspected journal version: 2007-12; 6872-6902 days (month precision)	inspected full text
F69: Panteleev-Kalachev: Degenerate Quantum LDPC Codes With Good Finite Length Performance Scalar-factor construction	first v1: 2019-04-04; 2760 days (day precision) inspected v3: 2021-11-12; 1807 days (day precision) journal milestone: 2021-11-22; 1797 days (day precision)	inspected full text
F70: Leverrier-Zemor: Quantum Tanner codes Constituent Ramanujan comparison	first v1: 2022-02-28; 1699 days (day precision) inspected v3: 2022-09-16; 1499 days (day precision) PDF internal date: 2022-09-19; 1496 days (day precision)	inspected full text
F71: Feng-Li: Spectra of Hypergraphs and Applications Classical source cited by Remark5.2; not a proof premise	publication metadata: 1996-09; 10981-11010 days (month precision)	abstract only full text gap
F72: Li-Sole: Spectra of Regular Graphs and Hypergraphs and Orthogonal Polynomials Classical comparison lead; not a proof premise	publication metadata: 1996; 10889-11254 days (year precision)	partial web text archival gap
F73: Prabhu et al.: On the Efficacy of the Peeling Decoder for the Quantum Expander Code Known stopping-set mechanisms and expansion conditions	first v1: 2025-04-30; 542 days (day precision) inspected v2: 2025-05-29; 513 days (day precision)	inspected full text
F74: Wang-Kulkarni-Poor: Finding All Small Error-Prone Substructures in LDPC Codes Classical rooted narrowing and branch-and-bound	related first preprint: 2006-05-19; 7463 days (day precision) inspected current-version date: 2009-04-22; 6394 days (day precision) journal issue: 2009-05; 6355-6385 days (month precision)	inspected full text

Source / role	Milestone date and elapsed calendar days	Access / precision limit
F75: Yao-Gokduman-Pfister: Cluster Decomposition for Improved Erasure Decoding of Quantum LDPC Codes B1 simulations / peeling-cluster comparison	first and inspected v1: 2024-12-11; 682 days (day precision)	inspected full text
Connolly2024: Connolly-Londe-Leverrier-Delfosse: Fast erasure decoder for hypergraph product codes Original stabilizer-support mechanism, credited through inspected F73/F75	first v1: 2022-08-01; 1545 days (day precision) v3 metadata: 2024-08-21; 794 days (day precision) journal publication: 2024-08-27; 788 days (day precision)	metadata only full text not consulted
F76: Prabhu-Gayathri-Kumar: On the Analysis of Stopping Sets Associated with the Quantum Hypergraph Product Code Later comparison lead; no exact theorem claim	conference metadata: 2026; 0-296 days (year precision)	full text access gap
RosnesYtrehus2007: Rosnes-Ytrehus: An Algorithm to Find All Small-Size Stopping Sets of Low-Density Parity-Check Matrices Bibliography-only enumeration lead	conference year: 2007; 6872-7236 days (year precision)	bibliography only not used as theorem evidence
Morris2025: Kirsten Morris: Analysis of Graph-based Decoders for Quantum Low Density Parity Check Codes Discovery trail through existing dissertation inventory	repository source version: 2025-08; 419-449 days (month precision)	discovery record not new full text screen
qLDPCOrg/qLDPC: qLDPCOrg/qLDPC pinned snapshot Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-23; 1 days (day precision)	pinned source or infrastructure
Lcrypto/Stopping-sets-enumerator: Lcrypto/Stopping-sets-enumerator pinned snapshot Software/infrastructure snapshot, not original publication age	inspected commit: 2023-10-29; 1091 days (day precision)	pinned source or infrastructure
leanprover-community/mathlib4: leanprover-community/mathlib4 pinned snapshot Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-21; 3 days (day precision)	pinned source or infrastructure
X-source-announcement: mathCObot source-paper announcement Accessible X comparison; not a mathematical answer	post UTC date: 2026-07-16; 100 days (day precision)	post inspected in parent audit
X-earlier-result: A_Lectionary earlier absorbing-set announcement Earlier repository result; different target	post UTC date: 2026-10-24; 0 days (day precision)	post inspected in parent audit

Source / role	Milestone date and elapsed calendar days	Access / precision limit
other-code-snapshots: Other inspected GitHub mirrors/README-only code Secondary discovery/control sources; no theorem or earliest-publication claim	recorded inspection: 2026-10-24; 0 days (day precision)	inspection date only earliest dates unassessed
Lean-runtime: Lean4.36.0-rc1 runtime Pinned proof checker; release-age claim not made	recorded use: 2026-10-24; 0 days (day precision)	use date only release date unassessed

Secondary mirrors and README-only sources have inspection-date deltas only; original publication dates remain unassessed. Lean release date likewise not verified; pinned runtime use is recorded.

The earliest date in a row means the earliest relevant milestone established by this audit, not a claim of historical priority. Tanner’s manuscript reports an older1982 submission, which is not treated as a verified public1982 publication. Metadata-only and unavailable sources supply no theorem premise and no zero-question screening claim. The X paper-announcement timestamp is16July2026UTC;15July is the arXiv paper date. Source age is distinct from the duration of an unresolved question.

The clause-by-clause audit and machine-readable date evidence are published at SKILL-COMPLIANCE-AUDIT.md and compliance-data.json. Existing named full-text, citation-index and social-search access gaps remain explicitly unresolved. Neither X posting nor a historical-priority claim is made.